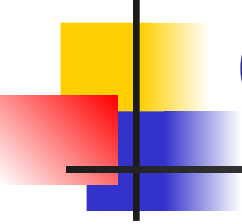


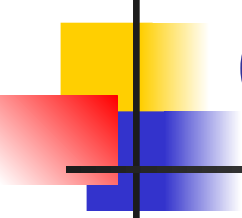
Python Coding

Dr. Bernard Chen
Troy University



Outline

- Intro
- Numbers
- Strings
- Lists



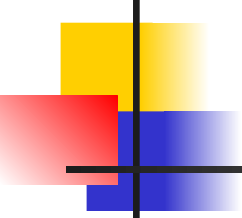
Introduction to Computer Science

- What do Computing Professionals Do?
 - We write program
 - Developing compute applications that address a need in some activity we humans do.



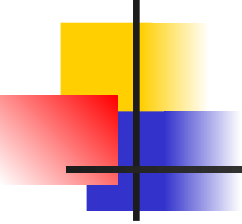
For example

Activity	Computer Application
Game	
Driving	GPS-based navigation software
Bank	Financial tools for investment
Shoppin g	Recommender system that suggest products
Films	3D computer graphics



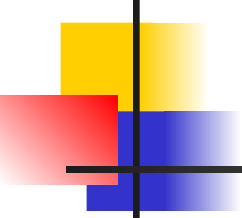
Program

- What distinguishes computer and other machines is that computers can be programmed
- The instructions that are actually executed inside computer is using binary notation (a sequence of 0s and 1s)
- There are many programming languages out there. Such as C++, C#, JAVA, Perl, and Python.



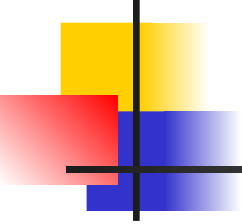
Python

- In this semester, we introduce Python programming language and use it to illustrate core computer science concepts and learn programming



Why do people use Python?

- Software Quality
- Developer productivity
- Program portability
- Support Libraries
- Component integration



Why do people use Python?

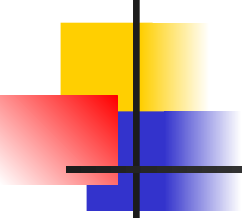
- **Software Quality:**

Python is designed to be readable, and hence maintainable.

Python is deep support for software reuse mechanisms such as OO.

- **Developer productivity:**

Python code is typically 1/3 to 1/5 the size of equivalent C++ or JAVA code



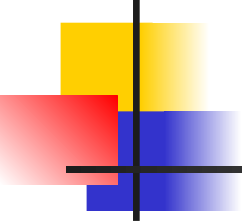
Why do people use Python?

- Program portability

Most python programs run unchanged on all major computer platforms

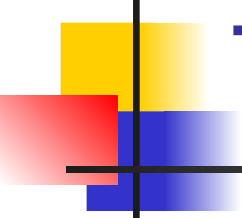
- Support Libraries
- Component integration

Today, Python code can invoke C and C++ libraries, can be called from C and C++, can integrate with Java components. Can communicate over XML, Corba and .NET etc



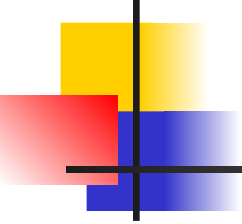
What can I do with Python?

- System Programming
- GUIs
- Internet Scripting
- Database Programming
- Games, Images, AI, XML and more



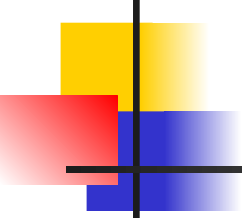
What are Python's Technical Strength

- It's OO (Object Oriented)
- It's free
- It's Portable
- It's Powerful
- **It's Easy to use**
- **It's Easy to learn**



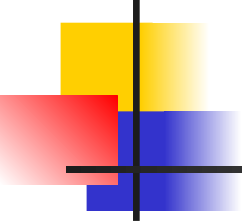
Hello World Program

- Implement by three different methods



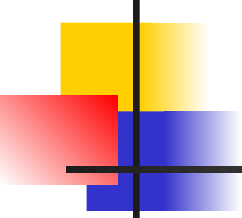
“Hello World” in C

```
main()  
{  
    printf("hello, world!\n");  
}
```



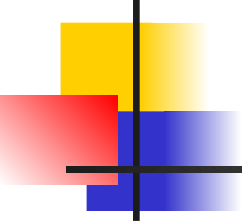
“Hello World” in JAVA

```
class myfirstjavaprogram
{
    public static void main(String args[])
    {
        System.out.println("Hello World!");
    }
}
```



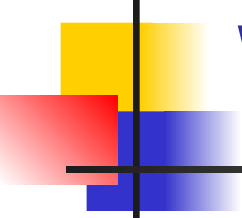
“Hello World” in Python

```
print ("hello World!")
```



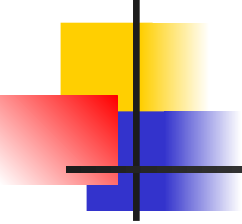
What is the Downside of Python?

- Perhaps the only downside to Python is that the execution speed may not always be as fast as compiled languages such as C and C++
- Python is not compiled all the way down to binary machine code, it is compiled to byte code instead.



Who Uses Python Today?

- Google and Yahoo currently use Python in Internet service
- IBM use Python for hardware testing
- Industrial Light and Magic use Python in the production of movie animation
- For more details, visit www.python.org



Install Python

- Go to <https://www.python.org/downloads/>
and download the latest version of
Python

Download Python | Python.org

python.org/downloads/

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Download Python 3.13.1

Looking for Python with a different OS? Python for [Windows](#), [Linux/UNIX](#), [macOS](#), [Other](#)

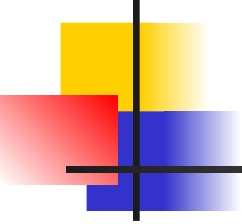
Want to help test development versions of Python 3.14? [Pre-releases](#), [Docker images](#)



Active Python Releases

For more information visit the [Python Developer's Guide](#).

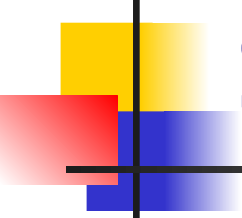
Python version	Maintenance status	First released	End of support	Release schedule
3.14	pre-release	2025-10-01 (planned)	2030-10	PEP 745
3.13	bugfix	2024-10-07	2029-10	PEP 719
3.12	bugfix	2023-10-02	2028-10	PEP 693



How do you run programs?

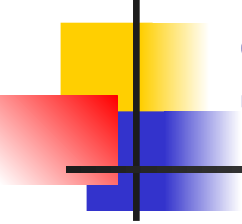
Three different methods:

- 1. Interactive Coding
- 2. Files (such as NotePad, WordPad)
- 3. Integrated Development Environment (IDE)



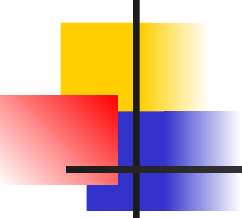
Some more Python codes

- `>>> 6+9`
- `>>> 6-9`
- `>>> 3/4`
- `>>> 3/4.0`
- `>>> 3**4`



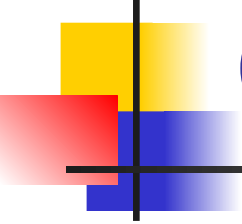
Some more Python codes

- `>>> print "Hello World!"`
- `>>> print 6+9`
- `>>> print "6+9"`



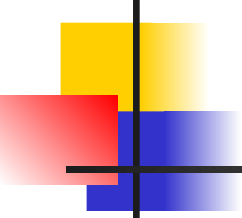
How do you run programs?

- Code in IDE
- Step 1. Select “new” in File
- Step 2. Select “Python Script” then click OK
- Step 3. Type in print "Hello World!"
- Step 4. Select “Save” in file (as “.py”)
- Step 5. Select “Run” (F5) in file



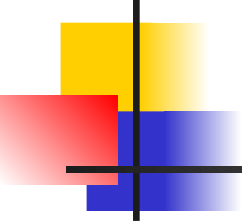
Outline

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Python Build-in types

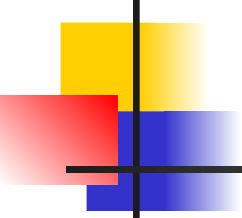
- Numbers 3.1415
- Strings “Hello World”
- Lists [1,2,3,4]
- Files
`input=open('file.txt', 'r')`



Numbers

Many different types, we only provide 3 here for now:

- Integer
- Floating-point
- Long integer

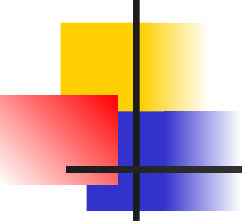


Number in Action

- Perhaps the best way to understand numerical objects is to see them in action.

```
>>> a=3      #Name Created
```

```
>>> b=4
```



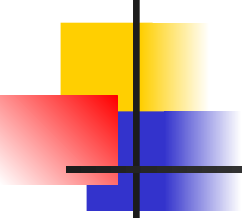
Number in Action

>>> a+1 # (3+1)

4

>>> a-1 #(3-1)

2



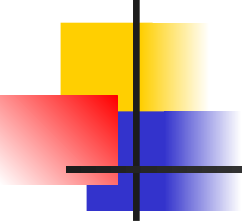
Number in Action

$>>> b*3$

12

$>>> b/2$

2



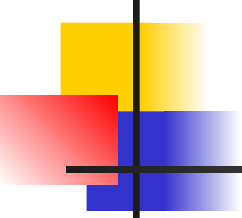
Number in Action

```
>>> b**2    #power
```

```
16
```

```
>>> a%2    #remainder
```

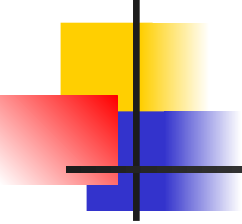
```
1
```



Number in Action

```
>>> 2+4.0, #mix type  
6.0
```

```
>>> 2.0**b  
16.0
```

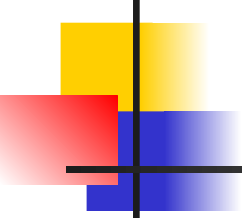


Number in Action

>>> b / 2 + a

>>> b/(2.0+a)

>>> b/(2+a)



Number in Action

```
>>> 1 / 2.0
```

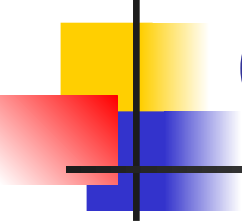
```
0.5
```

```
>>> 1/2
```

```
0.5
```

```
>>> int(1/2)
```

```
0
```



Other Numeric Tools

```
>>>int(2.567)
```

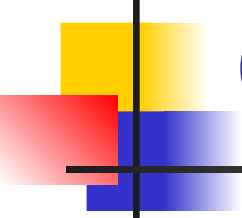
```
2
```

```
>>> round(2.567)
```

```
3.0
```

```
>>> round(2.567, 2)
```

```
2.57
```



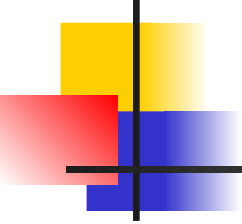
Other Numeric Tools

Math library

```
>>> import math  
>>> math.pi  
3.141592653589793
```

```
>>> math.e  
2.718281828459045
```

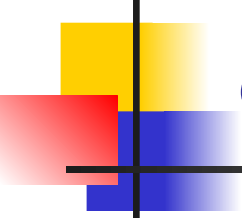
(in case you are interesting: <http://docs.python.org/library/math.html>)



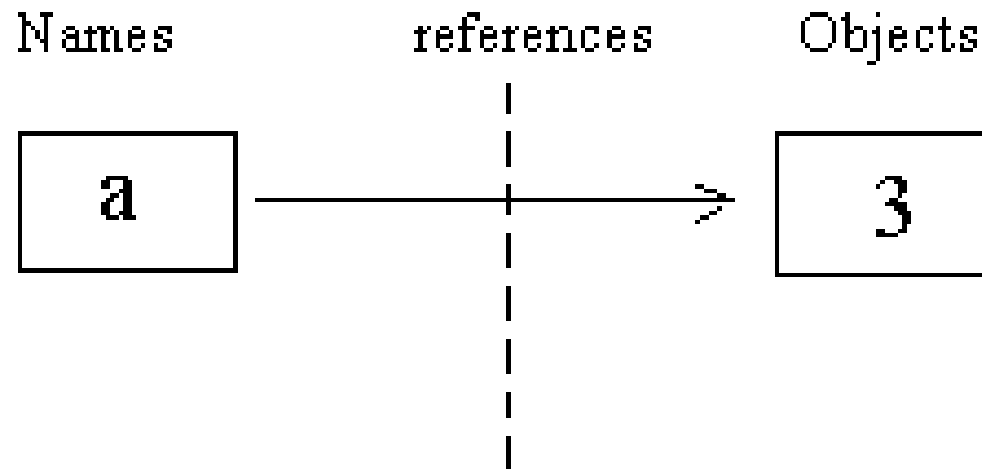
Dynamic Typing

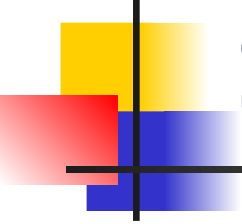
```
>>> a = 3
```

1. Create an object to represent the value of 3
2. Create the variable a, if it does not yet exist
3. Link the variable a to the new object 3



$a = 3$

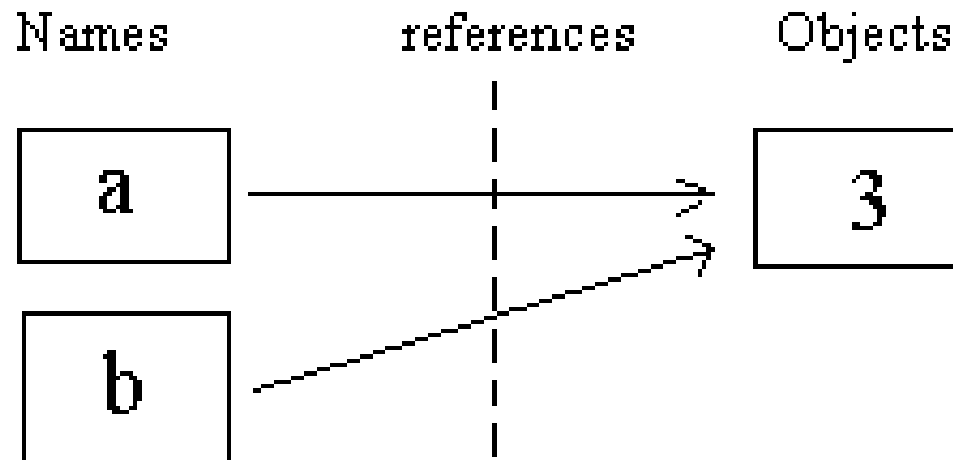




Share Reference

>>> a=3

>>> b=a

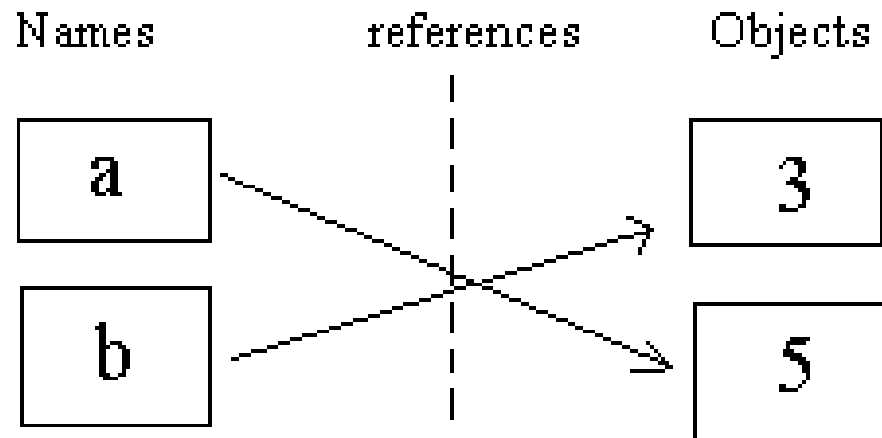


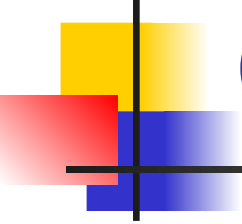
Share Reference

>>>a=3

>>>b=a

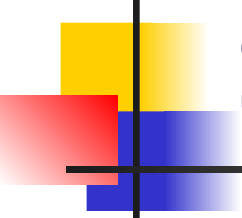
>>>a=5





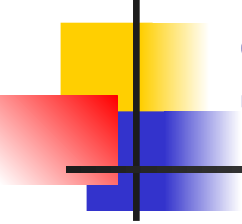
Outline

- Intro
- Numbers
- **Strings**
- Lists



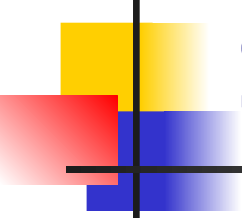
Strings

- The next major build-in type is the Python STRING --- an **ordered** collection of characters to store and represent text-based information
- Python strings are categorized as *immutable sequences* --- meaning they have a **left-to-right order** (sequence) and cannot be changed in place (immutable)



Single- and Double-Quoted

- Single- and Double-Quoted strings are the same
>>> 'Hello World' , "Hello World"
- The reason for including both is that it allows you to embed a quote character of the other inside a string
>>> "knight's" , 'knight"s'



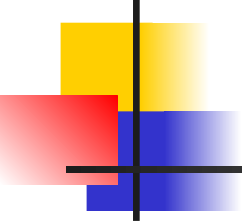
Strings in Action

- Basic operations:

1. `len()`

2. `+`

3. `*`



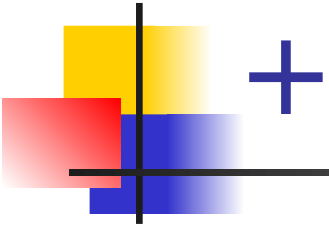
len()

- The len build-in function returns the length of strings

```
>>> len('abc')
```

```
>>> a='abc'
```

```
>>> len(a)
```



- Adding two string objects creates a new string object

```
>>> 'abc' + 'def'
```

```
>>> a='Hello'
```

```
>>> b='World'
```

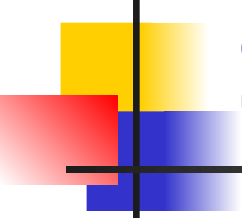
```
>>> a + b
```

```
>>> a+ ' ' +b
```



- Repetition may seem a bit obscure at first, but it comes in handy in a surprising number of contexts
- For example, to print a line of 80 dashes

```
>>> print ('_' * 80)
```



Strings in Action

- Advanced Operations
 1. Indexing
 2. Slicing
 3. String Conversion
 4. String Replace

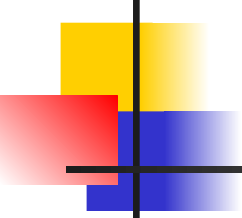
-

A diagram illustrating a Python string as an array of characters. The string "STRING IN PYTHON" is shown in a sequence of 14 boxes, each containing a character. Above the boxes are indices from 0 to 13. Below the first box is an upward arrow and the slice notation `[:]`, and below the last box is an upward arrow and the slice notation `[:]`.

0	1	2	3	4	5	6	7	8	9	10	11	12	13
S	T	R	I	N	G	I	N	P	Y	T	H	O	N

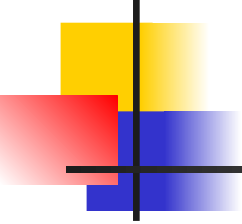
↑
[:]

↑
[:]



Indexing

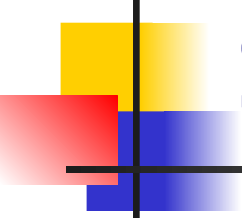
- As in the C language, Python offsests start at zero and end at one less than the length of the string
 - The first item is at offset 0
 - `S[0]` fetches the item at offsest 0 from the left
- >>> `S[0], S[5]`



Index

- Negative indexes mean to count backwards from the end
- `S[-2]` gets the item offset 2 from the end

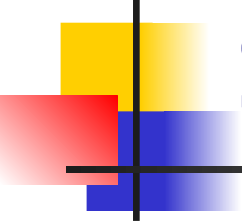
`>>> S[-2]`



Slicing

- Slicing allows us to extract an entire section (substring) in a single step

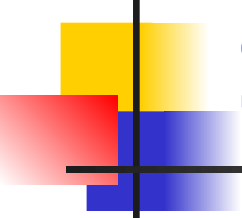
```
>>> S[1:3]    #extract item at 1 and 2  
>>> S[1:]     #all items pass the first  
>>> S[:-1]    #fetch all but the last  
item
```



Slicing

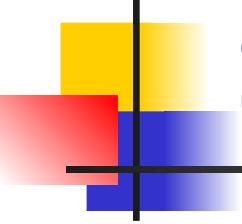
- You can regard the slicing function as

Start from	End At
[	]
—	—
:	
include	exclude



Slicing

- `S[1:3]`
- `S[1:]`
- `S[:3]`
- `S[:-1]`
- `S[:]`



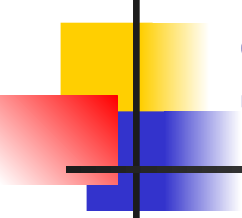
String Conversion

- You cannot add a number and a string together

```
>>> "42" + 1
```

- You can use `int()` function to convert string into integer

```
>>> int("42")
```

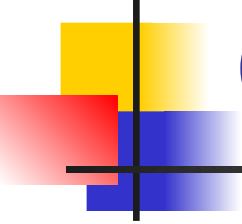


String Conversion

- Therefore, you can force adding one string to another number, and vise versa

```
>>> int("42") + 1
```

```
>>> "42" + str(1)
```

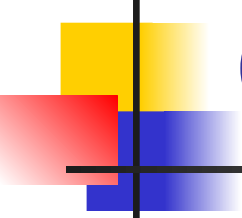


Changing Strings

- Remember the term “immutable”?
- It means that you cannot change a string in-place

```
>>> S = 'Spam'
```

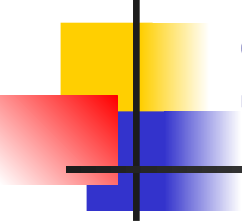
```
>>> S[0] = 'X'
```



Changing Strings

- There's a function called `replace()`

```
>>> S= 'string in python'  
>>> s.replace( 'string', 'number')
```



Escape Sequence Code Special Bytes

- `\n` --- Newline (table 5-2)
- `\t` --- Horizontal Tab

```
>>> s='a\nb\tc'
```

```
>>> s
```

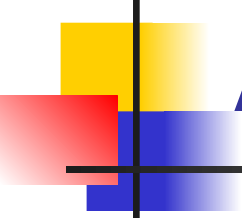
```
'a\nb\tc'
```

```
>>> print s
```

```
a
```

```
b
```

```
c
```



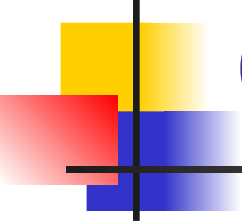
Assignment

- In the interactive shell, define:

```
>>> s1='-'
```

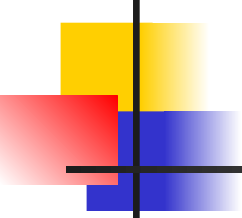
```
>>> s2='+'
```

- Now, use s1 and s2 and string operand + and * to present:
 1. `'-+'`
 2. `'+--'`
 3. `'+---+--'`
 4. `'+-++++--+-++++--+-++++--' (do this one)`



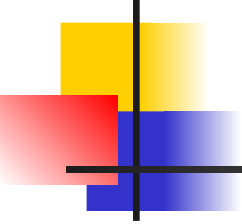
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- Strings
- **Lists**



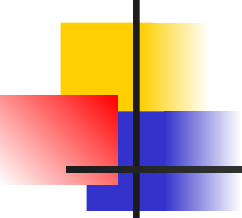
Lists

- Lists are Python's most flexible **ordered** collection object type
- Lists can contain any sort of object: numbers, strings and even other lists



Lists

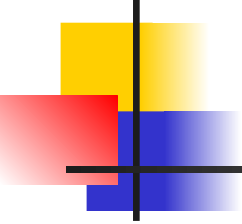
- **Ordered collections of arbitrary objects**
 - from the functional view, lists are just a place to collect other objects
 - Lists also define a left to right positional ordering of the items in the list



Lists

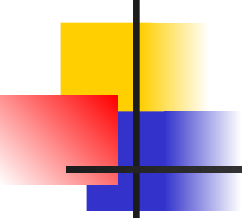
- **Accessed by offset**

- Just as strings, you can fetch a component object out of a list by indexing the list on the object's offset
- you can also for such tasks as slicing and concatenation



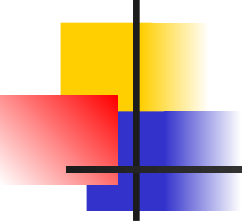
Lists

- **Variable length, heterogeneous, arbitrary nestable**
 - Unlike strings, list can grow and shrink in place
 - lists may contain any sort of object, not just one-character strings (they are heterogeneous)



Lists

- `>>> aa=[]` #An empty list
- `>>> aa=[1,2,3,4]` #4 items, index 0-3



List in action

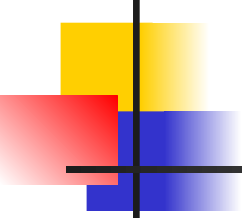
- List respond to the + and * operations much like strings

```
>>> aa=[1,2,3]
```

```
>>> bb=[4,5,6]
```

```
>>> aa+bb
```

```
>>> aa*3
```

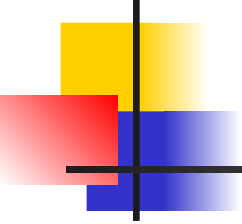


List in action

- Lists also have the function `len()` to tell the size of lists and “in” function

```
>>> aa=[1,2,3]
```

```
>>> len(aa)
```



Indexing, Slicing, and Matrixes

- Besides lists are sequences, index and slicing work the same way

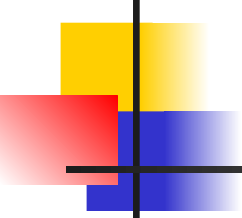
```
>>> aa=[1,2,3]
```

```
>>> aa[2]
```

```
>>> aa[-2]
```

```
>>> aa[1:]
```

(also try some strings in aa)



Indexing, Slicing, and Matrixes

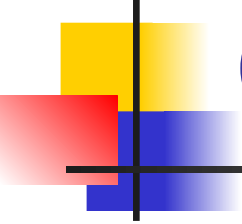
- We can also assign lists in a list

```
>>> matrix=[[1,2,3],[4,5,6],[7,8,9]]
```

```
>>> matrix[0]
```

```
>>> matrix[1][0]
```

```
>>> matrix[0][2]
```



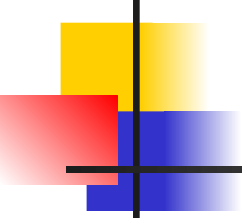
Changing Lists in-place

- When using a list, you can change its contents by assigning to a particular item (offset), or an entire section (slice)

```
>>> aa=[1,2,3]
```

```
>>> aa[0]=4
```

```
>>> aa[1:]=[5,6]
```



List method calls

The list append method simply tacks a single item onto the end of the list

```
>>> aa=[]
```

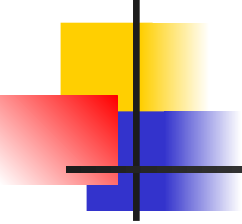
```
>>> aa.append(1)
```

```
>>> aa.append(2)
```

```
>>> aa.append(3)
```

```
>>> aa.append('4')
```

```
>>> aa.append(5.0)
```



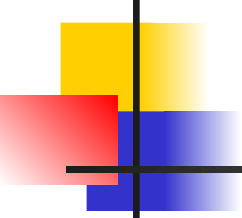
List method calls

- The sort function orders a list in place (in ascending fashion)

```
>>> aa=[4,2,6,8,1,3,4,10]
```

```
>>> aa.sort()
```

(also try strings in lists)

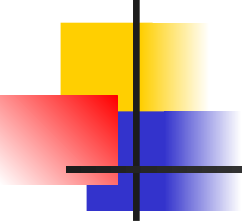


List method calls

- 'reverse' reverse the list in-place

```
>>> aa=[1,2,3,4]
```

```
>>> aa.reverse()
```



List method calls

- 'del' can be used to delete an item or section

```
>>> aa=[1,2,3,4,5,6]
```

```
>>> del aa[0]
```

```
>>> del aa[2:]
```